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4 / 4 submitted

Q1. String Character Frequency Counter

CODE

```
s=input("enter string:").lower()
freq={}
for ch in s:
    if ch != " ":
        freq[ch] = freq.get(ch,0) + 1
for ch in sorted(freq):
    print(ch, "→",freq[ch])
```

INPUT

Hello World

OUTPUT

```
enter string:h -> 1
e -> 1
h -> 1
e -> 1
h -> 1
l -> 1
e -> 1
h -> 1
l -> 2
e -> 1
h -> 1
l -> 2
o -> 1
-> 1
e -> 1
h -> 1
l -> 2
o -> 1
-> 1
e -> 1
h -> 1
l -> 2
o -> 1
w -> 1
-> 1
e -> 1
h -> 1
l -> 2
o -> 2
w -> 1
-> 1
e -> 1
h -> 1
l -> 2
o -> 2
r -> 1
w -> 1
-> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
-> 1
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics**CODE**

```

s=input().lower()
v="aeiou"
vf={}
cf={}
vc=cc=0
for i in s:
    if i in v:
        vc+=1
        vf[i]=vf.get(i,0)+1
    elif i.isalpha():
        cc+=1
        cf[i]=cf.get(i,0)+1
    print("vowels:",vc,"consonants:",cc)
    print("Most frequent vowel:",max(vf,key=vf.get))
    print("least frequent consonant:",min(cf,key=cf.get))
    print("ratio:",round(vc/cc,2))

```

INPUT

Python programming

OUTPUT

```

Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192420031RD_6a61d4e1ce7d04.80334085\main.py", line 14, in <module>
    print("Most frequent vowel:",max(vf,key=vf.get))
                                ^^^^^^^^^^^^^^^^^^^^^
ValueError: max() arg is an empty sequence

```

Q3. String Compression and Run-Length Encoding**CODE**

```

s=input()
c=""
i=0
while i < len(s):
    count=1
    while i+1 < len(s) and s[i] == s[i+1]:
        count += 1
        i += 1
    c += s[i] + str(count)
    i += 1
if len(c) < len(s):
    print("Compressed:",c)
else:
    print("Original:",s)
d = ""
i = 0
while i < len(c):
    ch = c[i]
    count = int(c[i+1])
    d += ch * count
    i += 2
print("Decompressed:", d)

```

INPUT

aabbccdddee

OUTPUT

```

Compressed: a2b3c2d4e2
Decompressed: aa
Decompressed: aabb
Decompressed: aabbcc
Decompressed: aabbccddd
Decompressed: aabbccdddee

```

7/27/26, 9:10 AM

Student Submissions — CO2_AT3_Coding Challenge

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Q4. Advanced Text Statistics Analyser

CODE

```
t=input().lower()
w=t.replace(".", "").replace("!", "").replace("?", "").split()
s=t.count(".") + t.count("!") + t.count("?")
if s==0: s=1
print("Words:", len(w))
print("Sentences:", s)
print("Paragraphs: 1")
print("Avg word length:", round(sum(len(i) for i in w)/len(w), 2))
print("Avg sentence length:", round(len(w)/s, 2))
f={}
for i in w:
    if i not in ["the", "is", "a", "an", "in", "of", "and", "to"]:
        f[i] = f.get(i, 0) + 1
print("Top 5 words:")
for i in sorted(f, key=f.get, reverse=True)[:5]:
    print(i, "(", f[i], ")")
```

INPUT

python is a powerful programming language

OUTPUT

```
Words: 6
Sentences: 1
Paragraphs: 1
Avg word length: 6.0
Avg sentence length: 6.0
Top 5 words:
python ( 1 )
powerful ( 1 )
programming ( 1 )
language ( 1 )
```